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Folding mechanism and folding electronic device

Abstract

A folding mechanism is disclosed. A first connecting portion is connected to a first rotating shaft, a second connecting portion is connected to a second rotating shaft, the first and second rotating shaft are rotatably connected to a first elastic connector, a first gear is mounted on the first rotating shaft, a second gear is mounted on the second rotating shaft and meshed with the first gear, the first gear has a first distal edge and a first proximal edge, the second gear has a second distal edge and a second proximal edge, the first elastic connector is elongated or shortened with meshed rotation of the first gear and the second gear, and the folding mechanism has a first state in which the first distal edge is meshed with the second distal edge, and a second state in which the first proximal edge is meshed with the second proximal edge.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of PCT International Application No. PCT/CN2021/142603 filed on Dec. 29, 2021, which claims priority to Chinese Patent Application No. 202011616737.8, entitled “FOLDING MECHANISM AND FOLDING ELECTRONIC DEVICE” and filed with the China National Intellectual Property Administration on Dec. 30, 2020, which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

(1) This application relates to the technical field of communication devices, and in particular, to a folding mechanism and a folding electronic device.

BACKGROUND

(2) With the development of technology, the development of electronic devices is increasingly rapid, and requirements of users for electronic devices are increasingly high. At present, a flexible screen is also widely applied to electronic devices, to form folding electronic devices.

(3) The folding electronic devices often have folding problems during folding. For example, when a folding electronic device is folded in, because a bending radius at a hinge is small, it is likely to cause crease or damage to the flexible screen by excessive extrusion. When the folding electronic device is folded out, because the bending radius at the hinge is excessive, it is likely to cause over pulling and deformation of the flexible screen, or even cause breaking of the flexible screen. It can be learned that, at present, the folding electronic device has a problem that the flexible screen is

easily damaged during folding, finally resulting in a short service life of the flexible screen.

SUMMARY

(4) According to a first aspect, an embodiment of this application discloses a folding mechanism, applied to a folding electronic device, including a first rotating shaft, a second rotating shaft, a first gear, a second gear, a first connecting portion, a second connecting portion, and a first elastic connector, where the first connecting portion is connected to the first rotating shaft, the second connecting portion is connected to the second rotating shaft, both the first rotating shaft and the second rotating shaft are rotatably connected to the first elastic connector, the first gear is mounted on the first rotating shaft, the second gear is mounted on the second rotating shaft, the first gear is meshed with the second gear, the first gear has a first distal edge and a first proximal edge, the second gear has a second distal edge and a second proximal edge, the first elastic connector is elongated or shortened with meshed rotation of the first gear and the second gear, and the folding mechanism has a first state in which the first distal edge is meshed with the second distal edge, and a second state in which the first proximal edge is meshed with the second proximal edge.

(5) According to a second aspect, an embodiment of this application discloses a folding electronic device, including a flexible screen, a first housing, a second housing, and the folding mechanism described above, where the first housing is connected to the first connecting portion and is rotatable with the first connecting portion, and the second housing is connected to the second connecting portion and is rotatable with the second connecting portion;

(6) the flexible screen includes a first region, a second region, and a joint region, the first region is connected to the second region through the joint region, the first region is connected to the first housing, and the second region is connected to the second housing;

(7) in a case that the first housing rotates relative to the second housing to a folded state of the electronic device, and the flexible screen is located between the first housing and the second housing, the first distal edge is meshed with the second distal edge; and in a case that the first housing rotates relative to the second housing to the folded state of the electronic device, and the first region and the second region are located on two opposite sides of the first housing and the second housing, the first proximal edge is meshed with the second proximal edge.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic structural diagram of a partial structure of an electronic device disclosed in an embodiment of this application when folded in;

(2) FIG. 2 and FIG. 3 are schematic structural diagrams of different parts of an electronic device disclosed in an embodiment of this application;

(3) FIG. 4 is a schematic structural diagram of a first occlusion structure of an electronic device disclosed in an embodiment of this application;

(4) FIG. 5 is a schematic structural diagram of a second occlusion structure of an electronic device disclosed in an embodiment of this application;

(5) FIG. 6 is a schematic structural diagram of a second elastic connector of an electronic device disclosed in an embodiment of this application;

(6) FIG. 7 is a schematic structural diagram of a first elastic mechanism or a second elastic mechanism of an electronic device disclosed in an embodiment of this application;

(7) FIG. 8 is a schematic diagram of a partial structure of an electronic device disclosed in an embodiment of this application;

(8) FIG. 9 is a schematic structural diagram of a partial structure of an electronic device disclosed in an embodiment of this application when folded in;

(9) FIG. 10 is a schematic structural diagram of a partial structure of an electronic device disclosed

in an embodiment of this application when folded out;

(10) FIG. 11 is a schematic structural diagram of a partial structure of an electronic device disclosed in an embodiment of this application when folded out.

LIST OF REFERENCE NUMERALS

(11) **100**—First rotating shaft, **110**—First occlusion structure, **111**—First occlusal teeth, **120**—Sleeve, **200**—Second rotating shaft, **210**—Second occlusion structure, **220**—Sliding rod, **300**—First gear, **310**—Second occlusal teeth, **311**—Fourth occlusal teeth, **400**—Second gear, **410**—Third occlusal teeth, **500**—First connecting portion, **510**—First limiting surface, **530**—First notch, **600**—Second connecting portion, **610**—Second limiting surface, **630**—Second notch, **700**—First elastic connector, **800**—First elastic mechanism, **900**—Second elastic mechanism, **1000**—Second elastic connector, **1100**—Flexible screen.

DETAILED DESCRIPTION

(12) The following clearly describes the technical solutions in the embodiments of this application with reference to the accompanying drawings in the embodiments of this application. Apparently, the described embodiments are some of the embodiments of this application rather than all of the embodiments. All other embodiments obtained by a person of ordinary skill in the art based on the embodiments of this application without creative efforts shall fall within the protection scope of this application.

(13) The specification and claims of this application, and terms “first” and “second” are used to distinguish similar objects, but are unnecessarily used to describe a specific sequence or order. It should be understood that the data in such a way are interchangeable in proper circumstances, so that the embodiments of this application can be implemented in other orders than the order illustrated or described herein. Objects distinguished by “first”, “second”, and the like are usually one type, and the number of objects is not limited. For example, the first object may be one or more than one. In addition, in the specification and the claims, “and/or” represents at least one of the connected objects, and the character “I” generally represents an “or” relationship between the associated objects.

(14) An electronic device provided in the embodiments of this application are described below through specific embodiments and application scenarios thereof with reference to the accompanying drawings.

(15) As shown in FIG. 1 to FIG. 11, an embodiment of this application discloses a folding mechanism, applied to a folding electronic device. The disclosed folding mechanism includes a first rotating shaft **100**, a second rotating shaft **200**, a first gear **300**, a second gear **400**, a first connecting portion **500**, a second connecting portion **600**, and a first elastic connector **700**.

(16) The first elastic connector **700** is a mounting base of the first rotating shaft **100** and the second rotating shaft **200**. Both the first rotating shaft **100** and the second rotating shaft **200** are rotatably connected to the first elastic connector **700**. The first elastic connector **700** may be a rubber tube, a silica sleeve, or the like. A specific type of the first elastic connector **700** is not limited in the embodiments of this application.

(17) Both the first rotating shaft **100** and the second rotating shaft **200** are rotatable about an axis of the first rotating shaft **100** and an axis of the second rotating shaft **200**. Optionally, a user may manually control the first rotating shaft **100** and the second rotating shaft **200** to rotate in a direction required by the user, to unfold or fold the folding electronic device.

(18) The first gear **300** is mounted on the first rotating shaft **100**, and during rotation of the first rotating shaft **100**, the first gear **300** rotates with the first rotating shaft **100**. The second gear **400** is mounted on the second rotating shaft **200**, and during rotation of the second rotating shaft **200**, the second gear **400** rotates with the second rotating shaft **200**. Optionally, the first gear **300** and the first rotating shaft **100** may be an integrated structure, and the second gear **400** and the second rotating shaft **200** may be an integrated structure. Certainly, the first gear **300** may be fixed to the first rotating shaft **100** in a key connection manner and the second gear **400** may be fixed to the

second rotating shaft **200** in the key connection manner. Specific assembling manners between the first gear **300** and the first rotating shaft **100** and between the second gear **400** and the second rotating shaft **200** are not limited in the embodiments of this application.

(19) The first gear **300** is meshed with the second gear **400** to ensure that the first rotating shaft **100** and the second rotating shaft **200** rotate synchronously. The first gear **300** has a first distal edge and a first proximal edge, and the second gear **400** has a second distal edge and a second proximal edge. The first distal edge of the first gear **300** is farthest from a rotation center of the first gear **300**, and the first proximal edge is closest to the rotation center of the first gear **300**. Similarly, the second distal edge of the second gear **400** is farthest from a rotation center of the second gear **400**, and the second proximal edge of the second gear **400** is closest to the rotation center of the second gear **400**.

(20) The first connecting portion **500** is connected to the first rotating shaft **100**, and the second connecting portion **600** is connected to the second rotating shaft **200**. The first connecting portion **500** is rotatable with the first rotating shaft **100**, and the second connecting portion **600** is rotatable with the second rotating shaft **200**. In a feasible operation mode, the user can manually rotate the first connecting portion **500** and the second connecting portion **600**, to implement rotation of the first rotating shaft **100** and the second rotating shaft **200** relative to the first elastic connector **700**, thereby implementing meshed rotation of the first gear **300** and the second gear **400**.

(21) The folding mechanism has a first state and a second state. In the first state, the first distal edge is meshed with the second distal edge, an axis center distance between the first rotating shaft **100** and the second rotating shaft **200** increases, the first elastic connector **700** is stretched, a bending radius increases, and the folding electronic device is finally in a folded state (a first folded state) with the increase of the axis center distance between the first rotating shaft **100** and the second rotating shaft **200**. In this case, the folding electronic device can be folded in, thereby avoiding crease or damage caused by excessive extrusion of the folding electronic device. In the second state, the first proximal edge is meshed with the second proximal edge, the axis center distance between the first rotating shaft **100** and the second rotating shaft **200** decreases, the first elastic connector **700** is squeezed, and the bending radius decreases. In this case, the folding electronic device may be folded out, so that the folding electronic device is finally in a folded state (a second folded state). In this process, since the bending radius decreases, the folding electronic device can be prevented from being over pulled or torn.

(22) The folding electronic device has an unfolded state. When bent in a first direction in the unfolded state, the folding electronic device can be folded out, and when bending in a second direction in the unfolded state, the folding electronic device can be folded in. The first direction is opposite to the second direction.

(23) The first elastic connector **700** may be elongated or shortened with meshed rotation of the first gear **300** and the second gear **400**. Optionally, when the first gear **300** and the second gear **400** are driven to be meshed by rotation of the first rotating shaft **100** and the second rotating shaft **200**, a distance between the first rotating shaft **100** and the second rotating shaft **200** may be changed with the change of the axis center distance between the first gear **300** and the second gear **400**, to further drive the first elastic connector **700** to be elongated or shortened. That is, the first elastic connector **700** adapts to the change of the axis center distance between the first gear **300** and the second gear **400** during meshing by elastic deformation of the first elastic connector **700**, thereby ensuring normal rotation and meshing.

(24) In the folding electronic device disclosed in the embodiments of this application, by improving the structure of the electronic device in the related art, the first gear **300** and the second gear **400** are mounted on the first rotating shaft **100** and the second rotating shaft **200** respectively, the first rotating shaft **100** is rotatably connected to the second rotating shaft **200** through the first elastic connector **700**, and under extension and compression of the first elastic connector **700**, the axis center distance between the first rotating shaft **100** and the second rotating shaft **200** changes. In a

case that the folding mechanism is in the first state, the first distal edge is meshed with the second distal edge, so that the axis center distance increases and the bending radius increases, thereby ensuring that the folding electronic device can be folded in a case of avoiding crease and damage caused by excessive extrusion of the folding electronic device. In a case that the folding mechanism is in the second state, the first proximal edge is meshed with the second proximal edge, so that the axis center distance decreases and the bending radius decreases, thereby ensuring that the folding electronic device can be folded out without being over pulled or broken. It can be learned that, in the folding electronic device disclosed in the embodiments of this application, by adjusting the axis center distance between the first rotating shaft **100** and the second rotating shaft **200**, the problem of the short service life of the flexible screen of the folding electronic device because the flexible screen is easily damaged during folding can be resolved.

(25) In the embodiments of this application, the first rotating shaft **100** is slidably sleeved with a first occlusion structure **110**, a surface of the first occlusion structure **110** facing the first gear **300** is provided with first occlusal teeth **111**, an end surface of the first gear **300** facing the first occlusion structure **110** is provided with second occlusal teeth **310**, and the first connecting portion **500** has a first limiting surface **510** facing the first gear **300**. Optionally, one end of a first elastic mechanism **800** may be positioned on the first limiting surface **510**, and the other end may be positioned on the first occlusion structure **110**. The first elastic mechanism **800** in a pre-pressed state is arranged between the first occlusion structure **110** and the first limiting surface **510**, the first occlusal teeth **111** are elastically meshed with the second occlusal teeth **310**, and the first gear **300** is in contact with the first elastic connector **700**; the first elastic mechanism **800** between the first occlusion structure **110** and the first limiting surface **510** may push the first occlusion structure **110**, so that the second occlusal teeth **310** and the first occlusal teeth **111** can maintain an meshed state, thereby facilitating hovering of the first connecting portion **500**, and finally implementing hovering of the folding electronic device at any angle during folding or unfolding.

(26) The second rotating shaft **200** is slidably sleeved with a second occlusion structure **210**, an end surface of a second gear **400** facing the second occlusal teeth **310** is provided with third occlusal teeth **410**, an end surface of the second occlusion structure **210** facing the second gear **400** is provided with fourth occlusal teeth **311**, and the second connecting portion **600** has a second limiting surface **610** facing the second gear **400**. Optionally, one end of a second elastic mechanism **900** may be positioned on the second limiting surface **610**, and the other end may be positioned on the second occlusion structure **210**. The second elastic mechanism **900** in a pre-pressed state is arranged between the second occlusion structure **210** and the second limiting surface **610**, the third occlusal teeth **410** are elastically meshed with the fourth occlusal teeth **311**, and the second gear **400** is in contact with the first elastic connector **700**. The second elastic mechanism **900** between the second occlusion structure **210** and the second limiting surface **610** may push the second occlusion structure **210**, so that the fourth occlusal teeth **311** and the third occlusal teeth **410** can maintain an meshed state, thereby facilitating hovering of the second connecting portion **600**, and finally implementing hovering of the folding electronic device at any angle during folding or unfolding.

(27) In this case, the first elastic mechanism **800** maintains elastic meshing of the first occlusal teeth **111** and the second occlusal teeth **310**, and the second elastic mechanism **900** maintains elastic meshing of the third occlusal teeth **410** and the fourth occlusal teeth **311**. The first elastic mechanism **800** and the second elastic mechanism **900** can maintain the first gear **300** and the second gear **400** at any meshed position, thereby ensuring that the folding mechanism can hover at will. The structure is beneficial to generating more obvious sense of mechanical operation when the user manually controls folding or unfolding, thereby improving sense of mechanical control of the user.

(28) In an optional solution, one of the first occlusion structure **110** and the second occlusion structure **210** may be provided with a sleeve **120**, the other is provided with a sliding rod **220**, and

the sliding rod **220** is in sliding fit with the sleeve **120**. In the process, the first occlusion structure **110** and the second occlusion structure **210** are slidably sleeved on the first rotating shaft **100** and the second rotating shaft **200** respectively, to rotate relative to the first rotating shaft **100** and the second rotating shaft **200**. However, the sliding rod **220** is sleeved inside the sleeve **120**, so that the first occlusion structure **110** and the second occlusion structure **210** do not rotate about axes of the first rotating shaft **100** and the second rotating shaft **200**, but may reciprocate in an extension direction of the sliding rod **220**, and the axis center distance between the first rotating shaft **100** and the second rotating shaft **200** changes while the sliding rod **220** slides. In this case, during folding of the folding electronic device, the axis center distance between the first rotating shaft **100** and the second rotating shaft **200** increases or decreases. In addition, the distance between the first occlusion structure **110** and the second occlusion structure **210** increases or decreases with sliding of the sliding rod **220** and the sleeve **120**, and the bending radius of the folding electronic device increases or decreases, so that the folding electronic device is smoother during folding, and the distance between the first rotating shaft **100** and the second rotating shaft **200** changes more accurately during rotation, thereby improving stability of folding deformation. A sliding fit direction between the sliding rod **220** and the sleeve **120** may be perpendicular to an axial direction of the first rotating shaft **100** or the second rotating shaft **200**.

(29) In a further technical solution, the first elastic mechanism **800** may be arranged between the first limiting surface **510** and one of the sleeve **120** and the sliding rod **220**, that is, one end of the first elastic mechanism **800** fits one of the sleeve **120** and the sliding rod **220**, and the other end of the first elastic mechanism **800** fits the first limiting surface **510**. The second elastic mechanism **900** may be arranged between the second limiting surface **610** and the other one of the sleeve **120** and the sliding rod **220**, that is, one end of the second elastic mechanism **900** fits the other one of the sleeve **120** and the sliding rod **220**, and the other end of the second elastic mechanism **900** fits the second limiting surface **610**.

(30) Optionally, the sleeve **120** and the sliding rod **220** may be mounted on the first occlusion structure **110** and the second occlusion structure **210** in two ways. In a case that the first occlusion structure **110** is provided with the sleeve **120** and the second occlusion structure **210** is provided with the sliding rod **220**, the first elastic mechanism **800** is arranged between the sleeve **120** and the first limiting surface **510**, and the second elastic mechanism **900** is arranged between the sliding rod **220** and the second limiting surface **610**; and in a case that the first occlusion structure **110** is provided with the sliding rod **220** and the second occlusion structure **210** is provided with the sleeve **120**, the first elastic mechanism **800** is arranged between the sliding rod **220** and the first limiting surface **510**, and the second elastic mechanism **900** is arranged between the sleeve **120** and the second limiting surface **610**. In this case, the sleeve **120** and the sliding rod **220** may be used to assist in mounting the first elastic mechanism **800** and the second elastic mechanism **900**, thereby implementing positioning of the first elastic mechanism **800** and the second elastic mechanism **900**.

(31) In an optional solution, a screw hole may be provided on a contact part between the first limiting surface **510** and the first rotating shaft **100**, threads are arranged on an outer contact surface between the first rotating shaft **100** and the first limiting surface **510**, and the first rotating shaft **100** is connected to the first connecting portion **500** through screw-threaded fit. In an optional solution, a screw hole is provided at a contact part between the second limiting surface **610** and the second rotating shaft **200**, and threads are arranged on an outer contact surface between the second rotating shaft **200** and the second limiting surface **610**, and the second rotating shaft **200** may be connected to the second limiting surface **610** through screw-threaded fit. Certainly, the first rotating shaft **100** and the first connecting portion **500**, and the second rotating shaft **200** and the second connecting portion **600** may be assembled in manners such as bonding, welding, and riveting.

(32) To make more standard elastic deformation of the first elastic mechanism **800** and the second elastic mechanism **900**, in an optional solution, the first elastic mechanism **800** may be sleeved on the first rotating shaft **100**. Similarly, the second elastic mechanism **900** may be sleeved on the

second rotating shaft **200**. The first elastic mechanism **800** and the second elastic mechanism **900** may be springs or other components capable of elastic deformation (for example, an elastic block or an elastic sheet), and specific structures of the first elastic mechanism **800** and the second elastic mechanism **900** are not limited in the embodiments of this application.

(33) In an optional solution, both the first elastic mechanism **800** and the second elastic mechanism **900** may be helical scaling springs.

(34) In this case, when the user folds the folding electronic device, the deformation of the first elastic mechanism **800** is changed through a change in a meshing state of the first occlusal teeth **111** and the second occlusal teeth **310**. The maintenance of the deformation of the first elastic mechanism **800** causes hovering and damping of the first rotating shaft **100**. The second elastic mechanism **900** and the first elastic mechanism **800** are stretched or compressed synchronously, and the second elastic mechanism **900** is always tightly connected to the second occlusion structure **210** and the second limiting surface **610**. The deformation of the second elastic mechanism **900** is changed through a change in a meshing state of the third occlusal teeth **410** and the fourth occlusal teeth **311**. The maintenance of the deformation of the second elastic mechanism **900** causes hovering and damping of the second rotating shaft **200**, and finally the first rotating shaft **100** and the second rotating shaft **200** synchronously hover and damp, thereby avoiding the damage to the folding electronic device caused by excessive force or sudden removal by the user.

(35) As described above, one end of the first elastic mechanism **800** may be fixed to the first limiting surface **510**, and the other end of the first elastic mechanism **800** may be fixed to the first occlusion structure **110**. One end of the second elastic mechanism **900** may be fixed to the second limiting surface **610**, and the other end of the second elastic mechanism **900** may be fixed to the second occlusion structure **210**. Therefore, in an optional solution, the first elastic mechanism **800** may be sleeved on the first rotating shaft **100**, or the second elastic mechanism **900** may be sleeved on the second rotating shaft **200**. In this case, the first elastic mechanism **800**, the first occlusion structure **110**, and the first limiting surface **510** are all arranged on the first rotating shaft **100**, and the second elastic mechanism **900**, the second occlusion structure **210**, and the second limiting surface **610** are all arranged on the second rotating shaft **200**. This arrangement is convenient for workers to assemble and reduces labor costs. In addition, neither the first limiting surface **510** nor the second limiting surface **610** is directly and fixedly connected to other components, so that the first limiting surface **510** and the second limiting surface **610** are not fixed stably with other components, thereby prolonging the service life of the folding electronic device.

(36) In another optional technical solution, the folding mechanism may further include a second elastic connector **1000**, and a material and a manufacturing process of the second elastic connector **1000** may be the same as those of the first elastic connector **700**, thereby reducing models during production and improving production efficiency. Both the first rotating shaft **100** and the second rotating shaft **200** are in sliding fit with the second elastic connector **1000**, the first gear **300** and the first elastic mechanism **800** are arranged between the first elastic connector **700** and the second elastic connector **1000**, and the second gear **400** and the second elastic mechanism **900** are arranged between the first elastic connector **700** and the second elastic connector **1000**. That is, the first gear **300** is located on one side of the first elastic connector **700** facing the first elastic mechanism **800**, and the first elastic mechanism **800** is located on one side of the second elastic connector **1000** facing the first gear **300**; and the second gear **400** is located on one side of the first elastic connector **700** facing the first elastic mechanism **800**, and the second elastic mechanism **900** is located on one side of the second elastic connector **1000** facing the second gear **400**. In this case, both the first elastic connector **700** and the second elastic connector **1000** can adapt to a change in the distance between the first rotating shaft **100** and the second rotating shaft **200** through deformation, and the first elastic connector **700** and the second elastic connector **1000** are spaced apart, thereby improving the balance of support. In addition, the first elastic connector **700** and the second elastic connector **1000** are spaced apart, which is more beneficial to maintaining a relative

angle between the first rotating shaft **100** and the second rotating shaft **200**. Generally, the first rotating shaft **100** and the second rotating shaft **200** are arranged in parallel, and in this case, an angle between the first rotating shaft **100** and the second rotating shaft **200** is zero.

(37) In an optional solution, the first elastic connector **700** and the second elastic connector **1000** are arranged at both ends of the first rotating shaft **100** and the second rotating shaft **200** respectively. When the user applies a force to the folding electronic device for folding, since both ends of the folding electronic device are arranged with elastic connectors in a folding direction, folding degrees of the both ends of the folding electronic device are the same during folding of the folding electronic device, thereby avoiding the damage of the folding electronic device caused by different folding degrees of the both ends during folding of the folding electronic device.

(38) In the embodiments of this application, the first connecting portion **500** may be fixedly connected to the first rotating shaft **100**, and the first connecting portion **500** may be provided with a first notch **530**. Optionally, the first rotating shaft **100** may be fixed to an edge of a length of the first connecting portion **500**, a diameter of the first rotating shaft **100** may be less than a thickness of the first connecting portion **500**, the first notch **530** may be provided at the edge of the length of the first connecting portion **500**, both ends of the first rotating shaft **100** are connected at two opposite edges of the first notch **530**, and the two edges of the first notch **530** may be drilled to fix the first rotating shaft **100** in holes at the two edges of the first notch **530**. The first rotating shaft **100** and the first notch **530** form a first accommodating space, and a part of the first gear **300** is located in the first accommodating space. Such a structure enables the first connecting portion **500** to sacrifice a partial structure of the first connecting portion **500** to form a space for accommodating at least a part of the first gear **300**, so that the folding mechanism is more compact in structure and occupies less space.

(39) The second connecting portion **600** may be fixedly connected to the second rotating shaft **200**, and the second connecting portion **600** may be provided with a second notch **630**. Optionally, the second rotating shaft **200** may be fixed to an edge of a length of the second connecting portion **600**, a diameter of the second rotating shaft **200** may be less than a thickness of the second connecting portion **600**, the second notch **630** may be provided at the edge of the length of the second connecting portion **600**, both ends of the second rotating shaft **200** are connected at two opposite edges of the second notch **630**, and the two edges of the second notch **630** may be drilled to fix the second rotating shaft **200** in holes of the two edges of the second notch **630**. The second rotating shaft **200** and the second notch **630** form a second accommodating space, and a part of the second gear **400** is located in the second accommodating space. Such a structure enables the second connecting portion **600** to sacrifice a partial structure of the second connecting portion **600** to form a space for accommodating at least a part of the second gear **400**, so that the folding mechanism is more compact in structure and occupies less space.

(40) In this case, the first accommodating space and the second accommodating space have a relative space, and at least part of the first gear **300** and the second gear **400** may be placed in the space, thereby reducing occupation of another space inside the folding electronic device, and being beneficial to reducing a size of the whole device and reducing material loss on the premise that all functions of the folding electronic device can be implemented.

(41) In the embodiments of this application, the first gear **300** may be a first convex gear, the second gear **400** may be a second convex gear, and the first convex gear and the second convex gear are not circular gears. During meshing of the first convex gear and the second convex gear, the distance between a rotation center of the first convex gear and a rotation center of the second convex gear changes, the first elastic connector **700** is stretched or compressed at the same time, the axis center distance between the first rotating shaft **100** and the second rotating shaft **200** also increases or decreases, and the bending radius of the folding electronic device also increases or decreases.

(42) Certainly, the first gear **300** may be a first eccentric gear, and the second gear **400** may be a

second eccentric gear. Compared with the convex gear, the eccentric gear is still a circular gear, but a rotation center of the eccentric gear is not a geometric center of the circular gear. Because the eccentric gear is still a circular gear, the eccentric gear has an advantage of simple manufacture.

(43) Based on the folding mechanism disclosed in the embodiments of this application, an embodiment of this application discloses a folding electronic device, which may include a flexible screen **1100**, a first housing, a second housing, and the folding mechanism described above.

(44) The flexible screen **1100** may be a light-emitting diode (Light-Emitting Diode, LED) flexible display screen, which has the advantages of strong flexibility, high resolution, and low power consumption. In addition, a mounting process is simple, and the flexible screen **1100** may be mounted by methods such as magnet adsorption and bonding, which simplifies the mounting and disassembling process, is also convenient for maintenance, and greatly reduces the mounting and maintenance costs.

(45) The first housing and the second housing may be metal housings or non-metal housings, and specific materials of the first housing and the second housing are not limited in the embodiments of this application. The first connecting portion **500** is connected to the first housing, and the second connecting portion **600** is connected to the second housing. During folding of the folding mechanism, the first housing and the second housing are rotatable relative to each other with relative rotation between the first connecting portion **500** and the second connecting portion **600**, thereby driving the flexible screen **1100** of the folding electronic device to be folded or unfolded.

(46) As described above, the first housing is connected to the first connecting portion **500**, and rotatable with the first connecting portion **500**, and the second housing is connected to the second connecting portion **600**, and rotatable with the second connecting portion **600**. The flexible screen **1100** includes a first region, a second region, and a joint region, the first region is connected to the second region through the joint region, the first region is connected to the first housing, and the second region is connected to the second housing. Optionally, the first region and the second region may be connected to surfaces of the first housing and the second housing in a bonding manner, and areas of surfaces of the first housing and the second housing bonded to the first region and the second region are the same as areas of the first region and the second region, to support the flexible screen **1100**.

(47) In a case that the first housing rotates relative to the second housing to a folded state of the folding electronic device and the flexible screen **1100** is located between the first housing and the second housing, the first distal edge is meshed with the second distal edge. Optionally, when the user does not use the folding electronic device, the flexible screen **1100** is meshed with the first gear **300** and the second gear **400** on the first rotating shaft **100** and the second rotating shaft **200**, the first distal edge of the first gear **300** is meshed with the second distal edge of the second gear **400**, the first elastic connector **700** and the second elastic connector **1000** are stretched, and the first elastic mechanism **800** and the second elastic mechanism **900** push the first occlusion structure **110** and the second occlusion structure **210** respectively to implement hovering and damping when the folding electronic device rotates. The axis center distance between the first rotating shaft **100** and the second rotating shaft **200** increases, and bending radiuses at the first connecting sheet and the second connecting sheet increase, so that the flexible screen **1100** of the folding electronic device is located between the first housing and the second housing, which can avoid direct contact between the flexible screen **1100** and an external environment, and avoid damage or friction of the flexible screen **1100**, thereby prolonging the service life of the flexible screen **1100**.

(48) In a case that the first housing rotates relative to the second housing to a folded state of the folding electronic device and the first region and the second region are located on two opposite sides of the first housing and the second housing, the first proximal edge is meshed with the second proximal edge. Optionally, when the user uses the electronic device, the flexible screen **1100** is meshed with the first gear **300** and the second gear **400** on the first rotating shaft **100** and the second rotating shaft **200**, the first proximal edge of the first gear **300** is meshed with the second

proximal edge of the second gear **400**, the first elastic connector **700** and the second elastic connector **1000** are compressed, and the first elastic mechanism **800** and the second elastic mechanism **900** push the first occlusion structure **110** and the second occlusion structure **210** respectively to implement hovering and damping when the folding electronic device rotates. The axis center distance between the first rotating shaft **100** and the second rotating shaft **200** decreases, so that the first region and the second region of the flexible screen **1100** of the folding electronic device are located on two opposite sides of the first housing and the second housing. Certainly, a suitable folding degree may be rotated by according to requirements of the user, so that the user can use the folding electronic device more comfortably according to preference of the user.

(49) In a further technical solution, the first connecting portion **500** may be a first connecting sheet fixedly connected to the first housing, and the second connecting portion **600** may be a second connecting sheet fixedly connected to the second housing. In an optional solution, the first connecting sheet and the first housing, and the second connecting sheet and the second housing may be connected in manners such as bonding, welding, and riveting. Materials of the first connecting sheet and the second connecting sheet may be sheet metal materials, and specific materials of the first connecting sheet and the second connecting sheet are not limited in the embodiments of this application.

(50) The folding and unfolding of the folding electronic device described above may be implemented through a manual operation. Certainly, the folding electronic device may further include a driving mechanism. The driving mechanism is connected to the first rotating shaft **100** or the second rotating shaft **200** to drive the first rotating shaft **100** or the second rotating shaft **200** to rotate, thereby implementing unfolding or folding of the folding mechanism. Compared with manual driving, mechanical driving is mainly reflected in that it is easier for the user when using the folding electronic device.

(51) The folding electronic device disclosed in the embodiments of this application may be a mobile phone, a tablet computer, an e-book reader, a game console, a wearable device, or the like. A specific type of the folding electronic device is not limited in the embodiments of this application.

(52) The foregoing embodiments of this application focus on describing differences between the embodiments. As long as different optimization features of the embodiments are not contradictory, the embodiments can be combined to form a better embodiment, and details are not described herein for the brevity of the text.

(53) The foregoing descriptions are merely embodiments of this application, and are not intended to limit this application. For a person skilled in the art, various modifications and variations may be made to this application. Any modification, equivalent replacement, or improvement made without departing from the spirit and principle of this application shall fall within the scope of the claims of this application.

Claims

1. A folding mechanism, applied to a folding electronic device, comprising a first rotating shaft, a second rotating shaft, a first gear, a second gear, a first connecting portion, a second connecting portion, and a first elastic connector, wherein the first connecting portion is connected to the first rotating shaft, the second connecting portion is connected to the second rotating shaft, both the first rotating shaft and the second rotating shaft are rotatably connected to the first elastic connector, the first gear is mounted on the first rotating shaft, the second gear is mounted on the second rotating shaft, the first gear is meshed with the second gear, the first gear has a first distal edge and a first proximal edge, the second gear has a second distal edge and a second proximal edge, the first elastic connector is configured to be elongated or shortened with meshed rotation of the first gear and the second gear, the folding mechanism has a first folding state in which the first distal edge is

meshed with the second distal edge, and a second folding state in which the first proximal edge is meshed with the second proximal edge.

2. The folding mechanism according to claim 1, wherein the first rotating shaft is slidably sleeved with a first occlusion structure, a surface of the first occlusion structure facing the first gear is provided with first occlusal teeth, an end surface of the first gear facing the first occlusion structure is provided with second occlusal teeth, the first connecting portion has a first limiting surface facing the first gear, a first elastic mechanism in a pre-pressed state is arranged between the first occlusion structure and the first limiting surface, the first occlusal teeth are elastically meshed with the second occlusal teeth, and the first gear is in contact with the first elastic connector; and/or the second rotating shaft is slidably sleeved with a second occlusion structure, an end surface of the second gear facing the second occlusal teeth is provided with third occlusal teeth, an end surface of the second occlusion structure facing the second gear is provided with fourth occlusal teeth, the second connecting portion has a second limiting surface facing the second gear, a second elastic mechanism in a pre-pressed state is arranged between the second occlusion structure and the second limiting surface, the third occlusal teeth are elastically meshed with the fourth occlusal teeth, and the second gear is in contact with the first elastic connector.

3. The folding mechanism according to claim 2, wherein one of the first occlusion structure and the second occlusion structure is provided with a sleeve, the other is provided with a sliding rod, and the sliding rod is in sliding fit with the sleeve.

4. The folding mechanism according to claim 3, wherein in a case that the first occlusion structure is provided with the sleeve and the second occlusion structure is provided with the sliding rod, the first elastic mechanism is arranged between the sleeve and the first limiting surface, and the second elastic mechanism is arranged between the sliding rod and the second limiting surface; and in a case that the first occlusion structure is provided with the sliding rod and the second occlusion structure is provided with the sleeve, the first elastic mechanism is arranged between the sliding rod and the first limiting surface, and the second elastic mechanism is arranged between the sleeve and the second limiting surface.

5. The folding mechanism according to claim 2, wherein the first elastic mechanism is sleeved on the first rotating shaft, or the second elastic mechanism is sleeved on the second rotating shaft.

6. The folding mechanism according to claim 2, wherein the folding mechanism further comprises a second elastic connector, both the first rotating shaft and the second rotating shaft are in sliding fit with the second elastic connector, the first gear and the first elastic mechanism are arranged between the first elastic connector and the second elastic connector, and the second gear and the second elastic mechanism are arranged between the first elastic connector and the second elastic connector.

7. The folding mechanism according to claim 1, wherein the first connecting portion is fixedly connected to the first rotating shaft, the first connecting portion is provided with a first notch, both ends of the first rotating shaft are connected to two opposite edges of the first notch, the first rotating shaft and the first notch form a first accommodating space, and a part of the first gear is located in the first accommodating space; and/or the second connecting portion is fixedly connected to the second rotating shaft, the second connecting portion is provided with a second notch, both ends of the second rotating shaft are connected to two opposite edges of the second notch, the second rotating shaft and the second notch form a second accommodating space, and a part of the second gear is located in the second accommodating space.

8. The folding mechanism according to claim 1, wherein the first gear is a first convex gear and the second gear is a second convex gear; or the first gear is a first eccentric gear and the second gear is a second eccentric gear.

9. A folding electronic device, comprising a flexible screen, a first housing, a second housing, and a folding mechanism, wherein the folding mechanism comprises a first rotating shaft, a second rotating shaft, a first gear, a second gear, a first connecting portion, a second connecting portion,

and a first elastic connector, the first connecting portion is connected to the first rotating shaft, the second connecting portion is connected to the second rotating shaft, both the first rotating shaft and the second rotating shaft are rotatably connected to the first elastic connector, the first gear is mounted on the first rotating shaft, the second gear is mounted on the second rotating shaft, the first gear is meshed with the second gear, the first gear has a first distal edge and a first proximal edge, the second gear has a second distal edge and a second proximal edge, the first elastic connector is configured to be elongated or shortened with meshed rotation of the first gear and the second gear, the folding mechanism has a first folding state in which the first distal edge is meshed with the second distal edge, and a second folding state in which the first proximal edge is meshed with the second proximal edge; the first housing is connected to the first connecting portion and is rotatable with the first connecting portion, and the second housing is connected to the second connecting portion and is rotatable with the second connecting portion; the flexible screen comprises a first region, a second region, and a joint region, the first region is connected to the second region through the joint region, the first region is connected to the first housing, and the second region is connected to the second housing; in a case that the first housing rotates relative to the second housing to a folded state of the electronic device, and the flexible screen is located between the first housing and the second housing, the first distal edge is meshed with the second distal edge; and in a case that the first housing rotates relative to the second housing to the folded state of the electronic device, and the first region and the second region are located on two opposite sides of the first housing and the second housing, the first proximal edge is meshed with the second proximal edge.

10. The folding electronic device according to claim 9, wherein the first connecting portion is a first connecting sheet fixedly connected to the first housing, and the second connecting portion is a second connecting sheet fixedly connected to the second housing.

11. The folding electronic device according to claim 9, wherein the first rotating shaft is slidably sleeved with a first occlusion structure, a surface of the first occlusion structure facing the first gear is provided with first occlusal teeth, an end surface of the first gear facing the first occlusion structure is provided with second occlusal teeth, the first connecting portion has a first limiting surface facing the first gear, a first elastic mechanism in a pre-pressed state is arranged between the first occlusion structure and the first limiting surface, the first occlusal teeth are elastically meshed with the second occlusal teeth, and the first gear is in contact with the first elastic connector; and/or the second rotating shaft is slidably sleeved with a second occlusion structure, an end surface of the second gear facing the second occlusal teeth is provided with third occlusal teeth, an end surface of the second occlusion structure facing the second gear is provided with fourth occlusal teeth, the second connecting portion has a second limiting surface facing the second gear, a second elastic mechanism in a pre-pressed state is arranged between the second occlusion structure and the second limiting surface, the third occlusal teeth are elastically meshed with the fourth occlusal teeth, and the second gear is in contact with the first elastic connector.

12. The folding electronic device according to claim 11, wherein one of the first occlusion structure and the second occlusion structure is provided with a sleeve, the other is provided with a sliding rod, and the sliding rod is in sliding fit with the sleeve.

13. The folding electronic device according to claim 12, wherein in a case that the first occlusion structure is provided with the sleeve and the second occlusion structure is provided with the sliding rod, the first elastic mechanism is arranged between the sleeve and the first limiting surface, and the second elastic mechanism is arranged between the sliding rod and the second limiting surface; and in a case that the first occlusion structure is provided with the sliding rod and the second occlusion structure is provided with the sleeve, the first elastic mechanism is arranged between the sliding rod and the first limiting surface, and the second elastic mechanism is arranged between the sleeve and the second limiting surface.

14. The folding electronic device according to claim 11, wherein the first elastic mechanism is sleeved on the first rotating shaft, or the second elastic mechanism is sleeved on the second rotating

shaft.

15. The folding electronic device according to claim 11, wherein the folding mechanism further comprises a second elastic connector, both the first rotating shaft and the second rotating shaft are in sliding fit with the second elastic connector, the first gear and the first elastic mechanism are arranged between the first elastic connector and the second elastic connector, and the second gear and the second elastic mechanism are arranged between the first elastic connector and the second elastic connector.

16. The folding electronic device according to claim 9, wherein the first connecting portion is fixedly connected to the first rotating shaft, the first connecting portion is provided with a first notch, both of the first rotating shaft are connected to two opposite edges of the first notch, the first rotating shaft and the first notch form a first accommodating space, and a part of the first gear is located in the first accommodating space; and/or the second connecting portion is fixedly connected to the second rotating shaft, the second connecting portion is provided with a second notch, both ends of the second rotating shaft are connected to two opposite edges of the second notch, the second rotating shaft and the second notch form a second accommodating space, and a part of the second gear is located in the second accommodating space.

17. The folding electronic device according to claim 9, wherein the first gear is a first convex gear and the second gear is a second convex gear; or the first gear is a first eccentric gear and the second gear is a second eccentric gear.
